Distributed Systems - Study Notes

1. CAP Theorem
A distributed data store can only guarantee two of the following three: Consistency, Availability and Partition tolerance. Because network partitions are unavoidable in a real cluster, the practical choice is between consistency and availability during a partition.

CP systems reject writes when a partition prevents quorum, which keeps every reader on the same value. AP systems keep serving reads and writes and reconcile conflicts later, usually with last-write-wins or CRDTs.

2. Consistency Models

Linearizability gives the illusion of a single copy updated atomically. Sequential consistency preserves each client's program order but allows reordering across clients. Causal consistency only orders operations that are causally related, which is enough for many collaborative editors.

3. Replication

Single-leader replication is simple to reason about but the leader is a bottleneck and a failover risk. Multi-leader replication tolerates regional outages at the cost of write conflicts. Leaderless replication uses quorums: with R + W > N, every read overlaps the latest write.

4. Consensus

Raft separates leader election, log replication and safety. A candidate wins an election with a majority of votes; each log entry carries a term number and an index. The commit rule only advances the commit index once a majority has replicated the entry from the current term.

5. Idempotency and Retries

Every retried request must carry an idempotency key so that a duplicated delivery cannot apply the same effect twice. Exponential backoff with jitter avoids synchronised retry storms.

6. TODO

TODO: compare Raft and Paxos commit rules in detail.
TODO: measure the write amplification of quorum replication.
TODO: write a failure-injection test for leader failover.